

Nakuja Project

Solid Propulsion Team

One-Grain Static Test — Dry Run Report

Document Version: 1.1 (Post-Execution Report)
Reference Procedure: Dry Run Procedure for One-Grain Static Test, v1.0
Date of Execution: Thursday, 9 July 2026
Location: Garden Space
Prepared by: Paschal Peters (Data Recorder / Timer)
Overall Outcome: **FAIL** (Procedural/Timeline) — **Ignition Successful**

1 Purpose

This report documents the execution of the One-Grain Static Test dry run conducted on 9 July 2026, in accordance with the *Dry Run Procedure for One-Grain Static Test (v1.0)*. The dry run was intended to validate the remote ignition circuit, team communication protocol, timing discipline, and overall readiness ahead of the live one-grain static test. This report records the sequence of events, faults encountered, corrective actions taken, and the final outcome, for reference in preparation for the repeat attempt scheduled the following morning.

2 Test Overview

Planned Start Time	08:00
Planned Duration	Approximately 3 hours
Total Attempts	3
Faults Recorded	2
Final Ignition Duration	6.84 seconds (ignition to termination)
Test Classification	Fail — exceeded allotted timeline despite eventual ignition success
Next Action	Repeat static test dry run scheduled for 8:00 am the following day

3 Sequence of Events

3.1 Attempt 1 — Scheduled 08:00, Actual Start 09:55

Delay Cause: Prior to commencing the sequence, the team carried out cross-checks and installed the meshed gate onto the protection cage housing the rocket motor, pushing the actual start to 09:55.

Fault Encountered: As the team prepared to begin, the Ignitions Team halted proceedings, reporting an issue. Upon inspection, the fault was traced to the **load cell battery**.

Resolution: The test was rescheduled to 14:00 to allow the battery issue to be addressed.

Fault #1	Load cell battery failure, identified during pre-ignition checks
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3.2 Attempt 2 — Scheduled 14:00, Actual Start 14:55

Delay Cause: The team conducted cross-checks, safety checks, and followed the established protocol prior to proceeding to the prime procedure.

Fault Encountered: Mid-installation, during the prime procedure, the ignition system activated on its own, without being triggered by the Ignition Operators. This was recorded as the second fault of the dry run.

Resolution: The team investigated the uncommanded activation, identified the cause, and corrected it before proceeding.

Fault #2	Uncommanded ignition system activation during prime procedure, opposite side from operator control
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3.3 Attempt 3 — Successful Execution

With both faults resolved, the team re-ran the full procedure without deviation. The sequence was executed cleanly with no issues observed.

Ignition to Termination Duration	6.84 seconds
Issues Observed	None
Result	Successful ignition sequence

4 Outcome Classification

Despite the successful ignition achieved on the third attempt, the dry run as a whole is classified as a **FAIL**. This classification is based on the operational timeline rather than ignition performance: the exercise was planned to take approximately 3 hours in total, but the combined delays from Fault #1 (load cell battery) and Fault #2 (uncommanded ignition activation) caused the total elapsed time to substantially exceed this allotment.

This distinction is intentional and consistent with the dry run's stated purpose — to identify issues *before* the live static test. The successful final ignition confirms the circuit and procedure are fundamentally sound; the timeline overrun confirms there are readiness gaps still to close.

5 Summary of Faults

Fault	Description	Attempt	Status
#1	Load cell battery failure detected during pre-ignition checks	Attempt 1 (09:55)	Resolved — rescheduled to 14:00
#2	Ignition system self-activated mid-installation during prime procedure, without operator trigger	Attempt 2 (14:55)	Resolved — inspected and corrected

6 Next Steps

- Both faults identified in this dry run are to be reviewed by the Ignitions Team and Safety Officer ahead of the repeat attempt.
- Load cell battery to be checked/replaced and verified prior to setup.
- Ignition circuit to be re-inspected for the cause of the uncommanded activation, with continuity and isolation re-verified before arming.
- A repeat static test dry run is scheduled for **8:00 am** the following morning, following the same procedure (v1.0).
- Emphasis for the repeat run: reducing setup-to-ignition time back within the 3-hour target by front-loading the cross-checks that caused delays in both Attempt 1 and Attempt 2.